

Tianrui Qi

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EDUCATION

Ph.D. in Biomedical Engineering

09/2025 - 12/2026

Boston University

B.S. in Computer Science

01/2023 - 05/2025

Georgia Institute of Technology

- GPA: 3.85/4.00
- Achievements: Highest Honor, President's Undergraduate Research Awards

B.S. in Computer Science, Mathematics

09/2020 - 12/2022

Rensselaer Polytechnic Institute

- GPA: 3.73/4.00
- Minor: Economics

PUBLICATIONS

†: contributed equally

Bio-CM²: Distributed Computational Optics for Cortex-Wide Cellular Imaging

07/2026

bioRxiv Preprint

Guorong Hu, Qilin Deng, Tianrui Qi, Zhixiong Chen, Bradley C Rauscher, Nathan Chai, Daria Bogatova, Bethany Weinberg, Jesse Smith, Ian G Davison, Martin Thunemann, Anna Devor, Lei Tian

DOI: doi.org/10.64898/2026.07.27.740823

Volumetric Localization Microscopy with Deep Learning

12/2025

Nature Communications

Keyi Han[†], Xuanwen Hua[†], Tianrui Qi[†], Zijun Gao, Xiaopeng Wang, Shu Jia

DOI: doi.org/10.1038/s41467-025-65941-3

Superiority of Quadratic over Conventional Neural Networks for Classification of Gaussian Mixture Data

09/2022

Visual Computing for Industry, Biomedicine, and Art

Tianrui Qi, Ge Wang

DOI: doi.org/10.1186/s42492-022-00118-z

RESEARCH EXPERIENCE

Graduate Research Assistant

11/2025 - 12/2026

Boston University

Computational Imaging Systems Lab, Advisor: Lei Tian, Ph.D.

- Analyzed reconstructed in vivo calcium-imaging videos covering a 10 mm × 7.5 mm field of view (FOV) from head-fixed mice to support biological validation of the computational miniature mesoscope (CM²).
- Applied Constrained Nonnegative Matrix Factorization for microendoscopic data (CNMF-E) to extract 3,474 neuronal footprints and corresponding 180-second calcium-activity traces from background-rich videos.
- Validated neuron extraction quality using spatial and temporal metrics, showing that the response kinetics of the GCaMP6s calcium indicator were consistent with published values.

GitHub: github.com/tianrui-qi/CM2-NeuronalSignal

PhD Rotation Student

03/2026 - 04/2026

Boston University

Dunlop Lab, Advisor: Mary Dunlop, Ph.D.

- Developed a physics-informed deep learning controller to predict light-stimulation schedules for target gene-expression trajectories, trained end-to-end using a differentiable ODE model of CcaS/CcaR dynamics.
- Characterized model performance across simulation stochasticity, input-history lengths, target-trajectory dynamics, and model architectures, identifying factors driving trajectory-tracking error.

GitHub: github.com/tianrui-qi/GEPC

PhD Rotation Student

09/2025 - 12/2025

Boston University

Biomedical Optical Technologies Lab, Advisor: Darren Roblyer, Ph.D.

- Developed a transformer-based cuffless blood pressure (BP) estimation pipeline driven by speckle contrast optical spectroscopy (SCOS), with flexible inputs handling variable-length and missing data.
- Established a three-stage training pipeline consisting of unsupervised representation learning on BFi and PPG waveforms, supervised regression on BP waveforms, and subject-specific finetuning.
- Visualized representations after unsupervised pretraining on optical waveforms, showing that subject-level differences were captured across measurements and that BP-related patterns were encoded.

GitHub: github.com/tianrui-qi/SCOS-BP

Undergraduate Research Assistant

04/2023 - 05/2025

Georgia Institute of Technology and Emory University

Jia Laboratory for Systems Biophotonics, Advisor: Shu Jia, Ph.D.

- Engineered a scalable 3D U-Net pipeline based entirely on simulated data for volumetric localization in single-molecule super-resolution microscopy, resolving sub-cellular structure down to 60 nm.
- Developed a patch-based prediction pipeline that flexibly adapts to various input volume sizes and achieves a 100x speedup over conventional deterministic localization methods.
- Integrated the redundant cross-correlation algorithm for drift correction with the deep learning-based prediction pipeline.

GitHub: github.com/tianrui-qi/DSMLM

Undergraduate Research Assistant

11/2021 - 12/2022

Rensselaer Polytechnic Institute

AI-based X-ray Imaging System Lab, Advisor: Ge Wang, Ph.D.

- Derived forward and backward propagation formulations for conventional and quadratic neural networks, with from-scratch NumPy implementation of full neural-network training workflows in Python.
- Demonstrated that single-layer quadratic neural networks rival conventional neural networks with hundreds of neurons in classifying simulated and real-world Gaussian mixture data.

GitHub: github.com/tianrui-qi/QuadraticNeurons

TEACHING EXPERIENCE

Graduate Teaching Assistant

09/2026 - 12/2026

Boston University

EC 522 Computational Optical Imaging, Instructor: Lei Tian, Ph.D.

Undergraduate Teaching Assistant

09/2022 - 12/2022

Rensselaer Polytechnic Institute

CSCI 2200 Foundations of Computer Science, Instructor: David Goldschmidt, Ph.D.

INDUSTRY EXPERIENCE

Co-op

01/2024 - 08/2024

Regeneron Genetics Center

Therapeutic Area Genetics, Manager: Jing He, Ph.D.

- Utilized BERT-based large language models (LLMs) and unsupervised feature selection to obtain a vector representation in a bio-meaningful space for each whole exome sequencing (WXS) sample.

- Demonstrated that the representations capture sample-wise differences by predicting immune system indicators of The Cancer Genome Atlas Program (TCGA) skin cancer samples.
- Scaled up the pipeline to handle hundreds of WXS samples with billion DNA sequences by optimizing parallel computing for high-performance computing (HPC) and enhancing file system efficiency through hashing.

GitHub: github.com/tianrui-qi/SIP-DB2

SKILLS

Programming: Python (PyTorch, Lightning, NumPy, pandas), MATLAB, Java, C, C++, R, Bash, MIPS.

Development Tools: Git, Conda, VS Code, RStudio, JetBrains Suite, Android Studio.

Computing Platforms: Linux (Ubuntu), AWS (EC2, S3), HPC (Slurm).

Software: Adobe Illustrator, ImageJ, LaTeX, OpenAI Codex.

Laboratory: optics and laser alignment, fluorescence imaging, fluorescence labeling, cell culture maintenance.

Communication: English (Professional), Mandarin (Native).

COURSE PROJECTS

Datapath and Control for a Turing Complete ISA with Interrupt Handling 09/2023 - 12/2023

Georgia Institute of Technology

CS 2200 Systems and Networking, Instructor: Daniel Forsyth

- Designed a single-bus datapath and an efficient four-ROM microcontroller for a Turing complete instruction set architecture (ISA), supporting arithmetic, logical, memory access, and control flow instructions.
- Handled basic and input device interrupts by additional hardware including new instructions, interrupt registers, signals, and I/O bus, along with software supports such as interrupt vector tables.

GitHub: github.com/tianrui-qi/LC2222a-ISA

Alternating Direction Method of Multipliers for Support Vector Machines 01/2022 - 05/2022

Rensselaer Polytechnic Institute

MATP 4820 Computational Optimization, Instructor: Yangyang Xu, Ph.D.

- Formulated the primal and augmented dual optimization problems for support vector machine (SVM) objective and developed alternating direction method of multipliers (ADMM) solver.
- Implemented the ADMM solver in MATLAB and reported the primal and dual feasibility violations at each outer iteration for the testing datasets.

GitHub: github.com/tianrui-qi/ADMM-for-SVM